



NORTH SYDNEY BOYS HIGH SCHOOL

**2010 Year 11
Assessment task 3**

Mathematics Extension 1

General Instructions

- Working time – **53 minutes**
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Mr Fletcher
- ☐ Mr Lam
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Name: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	Total	Total %
Mark	<u>12</u>	<u>14</u>	<u>14</u>	<u>17</u>	<u>57</u>	<u>100</u>

Question 1 (12 marks)

(a) Differentiate the following functions using the most appropriate method.

(i) $3x^5 + \frac{1}{x^2}$

2

(ii) $(13 + x^2)^7$

2

(iii) $x\sqrt{1-x^2}$

2

(b) Find

(i) $\lim_{x \rightarrow 0} \frac{x}{x^2 + 10}$

1

(ii) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x^2 - 4x}$

2

(c) The tangent to the curve $y = 2x^3 + x^2$ at $x = 1$ cuts the x -axis at $(a, 0)$. Find the equation of this tangent and the value of a .

3

Question 2 (14 marks) Start a new page.

(a) Differentiate from first principles: $y = f(x) = 5x^2 - 2x + 3$.

3

(b) For what values of x is the curve $y = x^3 - x^2 - 8x + 1$ decreasing?

3

(c) The size of classes (S) at a local public school is decreasing and the rate at which this is happening is decreasing.

i) Determine the sign of $\frac{dS}{dt}$ and $\frac{d^2S}{dt^2}$.

2

ii) Draw a graph of S versus t to show this information.

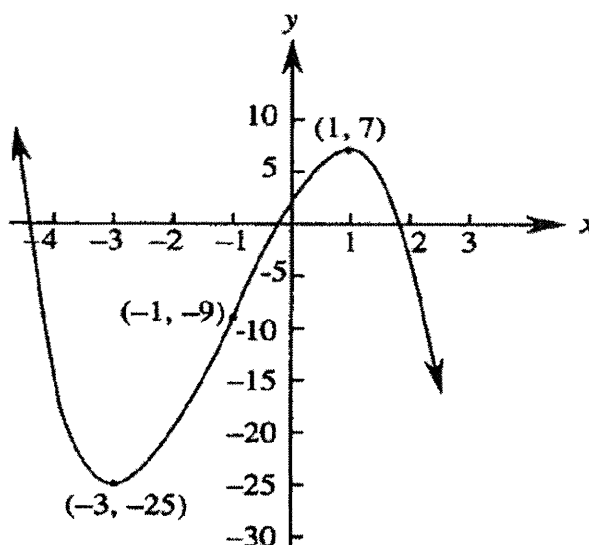
1

(d) For what values of b is the quartic $y = x^4 - bx^3 + 6x^2 + 13x - 7$ concave up at the point where $x = 2$?

3

(e) The curve $y = f(x)$ is shown below. The stationary points and point of inflexion are clearly shown. Sketch the graph of $y = f'(x)$ showing the x values of all important features.

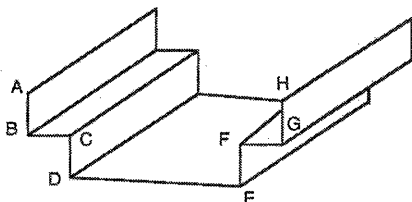
2



Question 3 (14 marks) Start a new page.

- (a) The surface area in cm^2 of a balloon being inflated is given by $S = t^3 - 2t^2 + 7t + 8$ where t is time in seconds. Find the rate of increase in the balloon's surface area after 5 seconds. 2
- (b) A particle moves in a straight line so that its displacement x metres from a fixed point O at time t seconds is given by $x = 3t^2 - 12t + 18$
- (i) Find its velocity and acceleration as a function of time. 2
- (ii) Find the initial velocity of the particle. 1
- (iii) Find when and where the particle is at rest. 2
- (iv) Find the distance travelled by the particle in the first 3 seconds. 2
- (c) Consider the function:
- $$f(x) = \begin{cases} \frac{1}{x} + 1 & x \leq -1 \\ \frac{x^2 - 1}{x - 1} & -1 < x < 2, \quad x \neq 1. \end{cases}$$
- (i) Is the function continuous at $x = -1$? Justify your answer. 2
- (ii) Is the function differentiable at $x = -1$? Justify your answer. 3

Question 4 (17 marks) Start a new page.

- (a) Find the point of inflexion on the curve $y = 2x^3 - 12x^2 + 5$ 3
- (b) If $y = f(x) = \frac{1+x^2}{1-x^2}$
- (i) Find any intercepts of the graph $y = f(x)$ with the y -axis. 1
- (ii) Find $\lim_{x \rightarrow \infty} \left(\frac{1+x^2}{1-x^2} \right)$ 1
- (iii) State the equation(s) of any asymptotes. 2
- (iv) Find the position and nature of any turning point(s). 3
- (v) Sketch the graph of $y = f(x)$ clearly showing all the above information 2
- (b) A metal gutter is to be folded as shown, from a strip 60 cm wide (i.e. The total distance ABCDEFGH is 60 cm.) AB, BC, CD, EF, FG and GH are each x cm and DE is y cm. All angles are right angles.
- 
- (i) Write a relation between x and y . 1
- (ii) Find the area of the cross section (DEFGHABCD) of the gutter in terms of x . 2
- (iii) Find the dimensions of the gutter if it is to have a maximum cross sectional area. 2

Solution of 2010 yr 11 Task 3 - Ext 1

marks.

Question 1:

$$(a) (i) y' = 15x^4 - 2x^{-3} \\ = 15x^4 - \frac{2}{x^3}$$

(i) ✓✓

$$(ii) 7(13+x^2)^6(2x) \\ = 14x(13+x^2)^6$$

 (ii) ✓✓ for correct rule.
0 for only $7(13+x^2)^6$

$$(iii) 1(\sqrt{1-x^2}) + x \cdot (-2x) \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \\ = \sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}} \\ = \frac{1-x^2-x^2}{\sqrt{1-x^2}} = \frac{1-2x^2}{\sqrt{1-x^2}}$$

 (iii) ✓ for $-\frac{x}{\sqrt{1-x^2}}$

✓ for correct product rule

$$(b) (i) \lim_{x \rightarrow 0} \frac{x}{x^2+10} = \frac{0}{10} = 0$$

(i) ✓

$$(ii) \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{x(x-4)} = \lim_{x \rightarrow 4} \frac{x+4}{x} \\ = \frac{8}{4} = 2$$

 (ii) ✓ for method.
✓ for 2.

$$(c) y' = 6x^2 + 2x \\ m = 6(1)^2 + 2(1) = 8$$

 (c) ✓ for $m=8$

$$x=1, y=2(1)^3+1^2=3$$

$$y-3=8(x-1)$$

$$y=8x-5 \quad \text{or} \quad 8x-y-5=0$$

$$\text{at } y=0: 8x-5=0$$

$$x=5/8=a$$

✓ correct line equation any f

 ✓ for $a=5/8$
Question 2:

a) First method:

$$f(x+h) = 5(x+h)^2 - 2(x+h) + 3 \\ = 5x^2 + 10xh + 5h^2 - 2x - 2h + 3$$

$$y' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + 5h^2 - 2x - 2h + 3 - 5x^2 + 2x - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{10x + 5h - 2}{1} = 10x - 2$$

(a)

 (Take away 1 if $\lim_{h \rightarrow 0}$ is not write

✓ correct rule

✓ correct method

✓ answer

Solution of 2010 yr 11 Task 3 - Ext 1
Question 2 cont'd:

2nd method: $y' = \lim_{u \rightarrow x} \frac{f(u) - f(x)}{u - x}$

$$y' = \lim_{u \rightarrow x} \frac{5u^2 - 2u + 3 - (5x^2 - 2x + 3)}{(u - x)}$$

$$= \lim_{u \rightarrow x} \frac{5(u^2 - x^2) - 2(u - x) + 3 - 3}{(u - x)}$$

$$= \lim_{u \rightarrow x} \frac{5(u+x)(u-x) - 2(u-x)}{(u-x)}$$

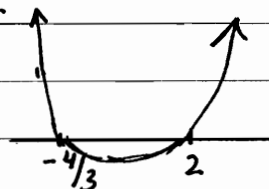
$$= \lim_{u \rightarrow x} [5(u+x) - 2] = 5 \times 2x - 2 = 10x - 2.$$

(b) $y = x^3 - x^2 - 8x + 1.$

$$y' = 3x^2 - 2x - 8 < 0.$$

$$(3x+4)(x-2) < 0.$$

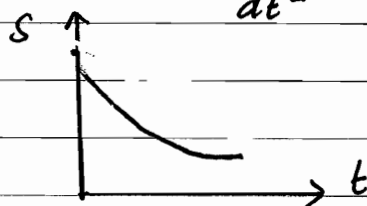
$$-4/3 < x < 2.$$



(c) (i) $\frac{ds}{dt} < 0$

$$\frac{d^2s}{dt^2} > 0.$$

(ii)



(d) $y' = 4x^3 - 3bx^2 + 12x + 13.$

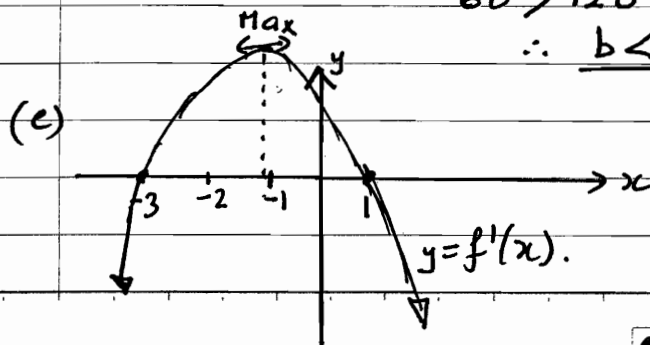
$$y'' = 12x^2 - 6bx + 12.$$

at $x=2$: $12(2)^2 - 6b(2) + 12 > 0.$

$$60 - 12b > 0.$$

$$60 > 12b$$

$$\therefore b < 5.$$



Marks.

or 2nd method.

✓ for correct principle with $\lim_{u \rightarrow x}$

✓ for correct method.

✓ for correct answer.

(b)

✓ for correct $y' < 0$

✓ for correct factors.

✓ for correct inequality.

(c)

(i) ✓ for $\frac{ds}{dt} < 0$

✓ for $\frac{d^2s}{dt^2} > 0$

(ii)

✓ for correct shape of graph

(d)

✓ for correct y'

✓ for $60 - 12b > 0$

✓ for $b < 5.$

(e)

✓ for correct shape.

✓ for all correct x values clearly shown $x = -3, -1, 1.$

Solution of 2010 Yr11 Task 3 - Ext1

Marks.

Question 3.

Q3-

(a) $S = t^3 - 2t^2 + 7t + 8.$

(a)

$$\frac{dS}{dt} = 3t^2 - 4t + 7.$$

✓

$$= 3(5)^2 - 4(5) + 7.$$

$$= 62 \text{ cm}^2/\text{s}.$$

✓ for $62 \text{ cm}^2/\text{s}.$

(b) $x = 3t^2 - 12t + 18.$

(b)

(i) $\frac{dx}{dt} = 6t - 12.$

(i) ✓ for $6t - 12$

$$\frac{d^2x}{dt^2} = 6 \text{ m/s}^2.$$

✓ for 6 m/s^2

(ii) at $t = 0$ $v = 6(0) - 12 = -12 \text{ m/s}.$

(ii) ✓ for $-12 \text{ m/s}.$
(not $12 \text{ m/s}.$)

(iii) $v = 6t - 12 = 0.$

(iii)

$$6t = 12 \quad \therefore t = 2 \text{ s}.$$

✓ $t = 2 \text{ s}.$

$$x = 3(2)^2 - 12(2) + 18.$$

$$x = 6 \text{ m}.$$

✓ for $x = 6 \text{ m}$

(iv)

t	0	2	3
x	18	6	9

(iv)

Distance travelled $0 \leq t \leq 2$ $x = 12 \text{ m}.$

$2 \leq t \leq 3$ $x = 3 \text{ m}.$

total distance: $12 + 3 = 15 \text{ m}.$

✓ for showing the 2 distance concept.

✓ for $15 \text{ m}.$

(c)

(i) $f(x) = \begin{cases} \frac{1}{x} + 1 & x \leq -1. \\ \frac{(x-1)(x+1)}{(x-1)} = x+1 & -1 < x < 2, x \neq 1. \end{cases}$

(c) students who write only: yes, no should only get 1 mark if both answers correct.

at $x = -1$: $\lim_{x \rightarrow -1^-} f(x) = \frac{1}{-1} + 1 = 0.$

$$\lim_{x \rightarrow -1^+} f(x) = -1 + 1 = 0.$$

✓ for correct working or even correct sketch.

$\therefore$ Continuous at $x = -1$ since $\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x).$

✓ for continuity.

(ii) $f'(x) = \begin{cases} -\frac{1}{x^2} \\ 1 \end{cases}$

(ii) (3 marks)

$$\lim_{x \rightarrow -1^-} f'(x) = -\frac{1}{(-1)^2} = -1.$$

$$\lim_{x \rightarrow -1^+} f'(x) = 1.$$

✓ for each correct derivative

$\therefore$ Not differentiable at $x = -1$ as



$\lim_{x \rightarrow -1^-} f'(x) \neq \lim_{x \rightarrow -1^+} f'(x).$ ✓ for not differentiable with the reason.

Solution of Ext1 Task 3 - 2010.

Question 4.

a) $y = 2x^3 - 12x^2 + 5.$

$y' = 6x^2 - 24x.$

$y'' = 12x - 24 = 0.$

$x = 2.$

test inflexion

x	1	2	3
y''	-12	0	+12
	concave down		concave up

$y = 2(2)^3 - 12(2)^2 + 5$
 $= -27.$

$\therefore (2, -27)$ is a point of inflexion since concavity changes.

(b) (i) at $x=0$, $y=1.$ (0,1)

(ii) $\lim_{x \rightarrow \infty} \frac{1+x^2}{1-x^2} = -1.$

(iii) $x = \pm 1$
 $y = -1$

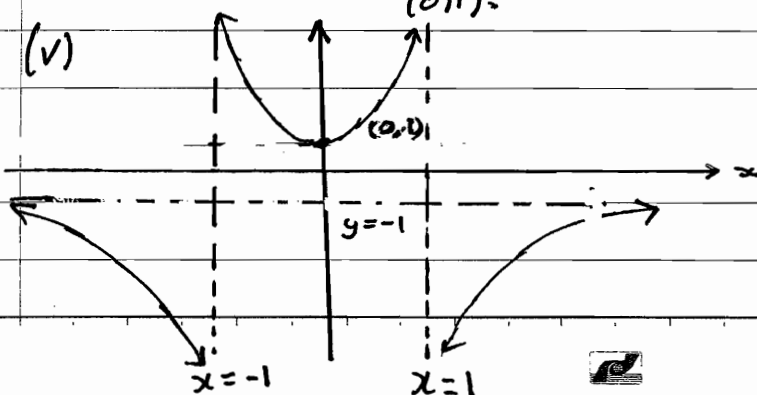
(iv) $y' = \frac{2x(1-x^2) - (-2x)(1+x^2)}{(1-x^2)^2}$
 $= \frac{2x - 2x^3 + 2x + 2x^3}{(1-x^2)^2}$
 $= \frac{4x}{(1-x^2)^2}$

$y' = 0 \therefore x=0, y=1$ (0,1)

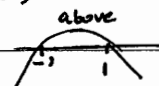
test:

	-2	0	2
$y' = \frac{4x}{(1-x^2)^2}$	-	0	+

(v)



sign diagram:
 $(1+x^2)(1-x)(1+x)$



a)

✓ for correct y'' .

✓ for test.

✓ for (2, -27).

(b)

(i) ✓ for $y=1$ or (0,1)

(ii) ✓ for -1.

(iii)

✓ $x = \pm 1$

✓ $y = -1$.

(iv)

✓ for correct y' (quotient rule)

✓ for (0,1)

✓ for testing.

(v)

✓✓ for correct shape with asymptote

Solution of Ext1 Task 3- 2010

Marks

(b)

$$(i) \quad 6x + y = 60.$$

(i) ✓

$$(ii) \quad A = xy + x(y + 2x)$$

(ii)

$$= xy + xy + 2x^2$$

$$= 2xy + 2x^2.$$

✓ for correct st
 $A = 2xy + 2x^2$

$$= 2x(60 - 6x) + 2x^2$$

$$= 120x - 12x^2 + 2x^2.$$

$$= 120x - 10x^2.$$

✓ for $120x - 10x^2$
(iii)

$$(iii) \quad \frac{dA}{dx} = 120 - 20x = 0.$$

$$120 = 20x.$$

✓ for correct x

$$x = 6 \text{ cm.}$$

$$y = 60 - 6x = 60 - 36 = 24 \text{ cm.}$$

✓ for $y = 24 \text{ cm}$
and testing
(both).

$$\frac{d^2A}{dx^2} = -20 < 0 \quad \therefore \text{Max.}$$

Follow through
errors in iii.
If area is wrong
they lose the
2 marks in (ii)